

New Zealand Math Olympiad Round 1 2022 — P3/8

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Find all real numbers x and y such that

$$\begin{aligned}x^2 + y^2 &= 2, \\ \frac{x^2}{2-y} + \frac{y^2}{2-x} &= 2.\end{aligned}$$

Solution

Manipulate the second equation to show that

$$\begin{aligned}x^2(2-x) + y^2(2-y) &= 2(2-x)(2-y) \\ 4 - (x^3 + y^3) &= 2(4 - 2(x+y) + xy) \\ 4(x+y) &= 4 + 2xy + (x^3 + y^3) \\ &= 2 + (x+y)^2 + (x^3 + y^3)\end{aligned}$$

Where the second line is due to $2(x^2 + y^2) = 4$, and the last line is due to $(x+y)^2 = x^2 + y^2 + 2xy = 2 + 2xy$. Now define $S := (x+y)$. Note that $S^3 = x^3 + y^3 + 3xyS$ and also since $S^2 = 2 + 2xy \iff xy = \frac{1}{2}S^2 - 1$ we can conclude that $S^3 = x^3 + y^3 + 3\left(\frac{1}{2}S^2 - 1\right)S$. Re-arranging and then substituting for $x^3 + y^3$ now yields

$$\begin{aligned}4S &= 2 + S^2 + \left(S^3 - 3\left(\frac{1}{2}S^2 - 1\right)S\right) \\ 0 &= S^3 - 2S^2 + 2S - 4 \\ 0 &= (S-2)(S^2+2)\end{aligned}$$

and since $S \in \mathbb{R}$ we must have $S = x + y = 2$. Thus $x^2 + (2-x)^2 = 2 \iff (x-1)^2 = 0$ and so $y = \pm 1$ from which it can easily be checked that the only solution is in fact $(x, y) = (1, 1)$. ■